

Kevin Du

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EDUCATION

University of Pennsylvania, School of Engineering and Applied Science, Philadelphia, PA

May 2027

B.S.E, DMD Computer Science and Computational Graphics Program

M.S.E, Computer Graphics and Game Technology

Cumulative GPA: 4.0

RELEVANT COURSEWORK

Data Structures and Algorithms, Machine Perception, Interactive Computer Graphics, Path Tracing and Physically Based Rendering, Procedural Computer Graphics & Design Systems, Computer Animation, Applied Machine Learning

EXPERIENCE

Penn Medicine, University of Pennsylvania Health System

May 2025 - August 2025

Virtual Reality Software Engineering Intern

Philadelphia, PA

- Developed a Virtual Reality Surgical Airway Training Simulator with Unity Engine to be run on Oculus Quest headsets
- Contributed to system design, debugging, and iterative testing
- Collaborated with surgical residents to create accurate replications of surgical procedure steps and ensure user-friendliness

UPenn Game Research and Development Environment

Sep 2023 - Present

Game Design & Project Lead

Philadelphia, PA

- Managed a team of 20+ people as the lead of the main 2024-2025 project published on Steam and Itch.io, responsible for implementing AI behavior systems and HLSL visual effects shaders as lead programmer
- Oversaw multiple teams in the design and development of the PennBoy minigame collection across successive years

Engineering Summer Academy at Penn

June - July 2024 & 2026

Residential Teaching Assistant

Philadelphia, PA

- Taught data structures, 3D design, and figure modeling techniques, providing hands-on assistance to high school students
- Planned and budgeted events and recreational activities

PROJECTS

3D Deferred Rendering Engine

C++, OpenGL, QT Creator

- Developed a real-time physically-based rendering engine with OpenGL to render photo-realistic images
- Utilizes PBR models, precomputed HDR image-based environment lighting, as well as post-process screen space reflection

Image2GPS

Python, PyTorch, Hugging Face, ResNet

- Built an end-to-end computer vision regression pipeline to predict campus-image GPS coordinates, collecting and curating a custom dataset of 1.6k labeled images
- Fine-tuned ResNet-based models with Smooth L1 loss, AdamW, learning-rate scheduling, and Haversine-distance validation

Game Development Projects

C#, C++, HLSL, Blender, Unity, Unreal Engine 5, Godot

- Led the development of **Catanks**, the 2024-2025 UPGRADE top-down arcade tank shooter; worked on game design planning, enemy AI behavior and pathfinding systems programming, VFX shaders, game asset pipeline coordination
- Currently the lead producer and dev of **UpgradeKart**, the upcoming 2025-2026 UPGRADE kart-racing game, in development

SKILLS

Languages: C++, C#, GLSL, HLSL, Java, Python, JavaScript, TypeScript, Swift, HTML/CSS

Frameworks and Tools: Unity, Unreal Engine 5, Git, CUDA, PyTorch, VS Code, QT Creator, Eclipse, OpenGL, Node.js, Jira